

VIJAY KUMAR

Curriculum Vitae

CONTACT

School of Computer Science
Data Science Cluster, UPES
Dehradun, Uttarakhand-248007

✉ dvijay.iitb@gmail.com

☎ (+91) 7905257340

Profile Links: Google Scholar; ResearchGate; AMS MathSciNet; LinkedIn;

ABOUT ME

Dedicated researcher at IIT Bombay, with a strong background in probability and statistics. I am exploring applications in statistics, stochastic processes and data science. Passionate about teaching, fostering growth and advancing statistical knowledge through collaboration and innovation.

EDUCATION

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|----------------------|---|
| Jan 2018 - Nov 2023 | Ph.D. in Statistics Department of Mathematics, Indian Institute of Technology, Bombay Maharashtra, India
Thesis Advisor: Prof. P. Vellaisamy
Thesis Title: Adomian Polynomials and some Fractional Stochastic Processes
CPI: 7.56/10 ($\equiv$ 75.6%) |
| July 2011 - May 2013 | M.Sc. in Applied Statistics and Informatics Department of Mathematics, Indian Institute of Technology, Bombay Maharashtra, India
CPI: 6.36/10 ($\equiv$ 63.6%) |
| July 2006 - May 2009 | B.Sc. in Statistics and Mathematics Department of Statistics, PPN Degree College, CSJM University, Kanpur Uttar Pradesh, India
Percentage: 66.74% |
| July 2004 - May 2005 | Intermediate UP Board Science Percentage: 75.80% |
| July 2002 - May 2003 | High school UP Board Science Percentage: 75.33% |

TEACHING AND RESEARCH EXPERIENCE

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|------------------------|---|
| January 2024 - Present | Assistant Professor School of Computer Science (Data Science Cluster) University of Petroleum and Energy Studies Dehradun, Uttarakhand, India. |
| July 2023 - Nov 2023 | Research Associate Department of Mathematics, Indian Institute of Technology, Bombay Maharashtra, India. |

RESEARCH AREA

Statistics, Probability, Stochastic process, Data Science, Distribution theory.

PUBLISHED PAPERS

1. KK Kataria, P Vellaisamy, V Kumar. *A probabilistic interpretation of the Bell polynomials* (2022). Stochastic Analysis and Applications, Taylor & Francis, 40(4), 610-622.
2. P Vellaisamy, V Kumar. *Probabilistic interpretations of nonclassic Adomian polynomials* (2022). Stochastic Analysis and Applications, Taylor & Francis, 40(5), 931-950.

3. M Arshad, M Khetan, V Kumar, AK Pathak. *Record-based transmuted generalized linear exponential distribution with increasing, decreasing and bathtub shaped failure rates* (2022). Communications in Statistics-Simulation and Computation, Taylor & Francis.
4. C Chesneau, V Kumar, M Khetan, M Arshad. *On a Modified Weighted Exponential Distribution with Applications* (2022). Mathematical and Computational Applications, 27(1), 17.
5. V Kumar, P Vellaisamy. *Three different derivations of the Poisson process* (2018). Mathematics Newsletter, 29(1), 24-28.

SUBMITTED PAPERS

1. V Kumar, A Chakraborty, M Arshad, A Tiwari. *A new Nadarajah-Haghighi distribution with applications* (2024). (under review).
2. M Khandakar, V Kumar, P Vellaisamy. *Extension of fractional Poisson process, birth process and death process* (2024). (under review).
3. M Arshad, KR Meena, V Kumar, M Khetan. *E-Bayesian estimation and its E-posterior risk for the exponentiated family of distributions based on generalized order statistics* (2024). (under review).

CHAPTERS IN EDITED BOOKS

Book: *Distribution Theory: Principles and Applications*. ISBN: 9781774912140, (Edited Book) Apple Academic Press, co-publishing with CRC Press, Taylor & Francis.

Editors: Fozia Homa, Mukti Khetan, Mohd. Arshad, Pradeep Mishra.

1. *Random Variables and Their Properties* (2023). **Vijay Kumar**, Mukti Khetan, Mohd. Arshad, and Shweta Dixit.
2. *Discrete Distributions* (2023). Mukti Khetan, **Vijay Kumar**, Mohd. Arshad, and Prashant Verma.
3. *Family of Weibull Distributions* (2023). Mohd. Arshad, **Vijay Kumar**, Mukti Khetan, and Fozia Homa.
4. *Descriptive Statistics: Principles and Applications* (2023). Fozia Homa, Mohd. Arshad, **Vijay Kumar**, Shubham Thakur, and Pradeep Mishra.
5. *Life Distributions* (2023). Mohd Arshad, Mukti Khetan, **Vijay Kumar**, and Fozia Homa.

TEACHING EXPERIENCE

I instruct diverse groups of students in various statistical courses. In my role as a teaching assistant at IIT Bombay, I got the opportunity to teach a range of courses, including Introduction to Probability and Statistics, Regression Analysis using R, Multivariate Analysis, Applied Stochastic Process, Parametric and Nonparametric Statistics. These courses have allowed me to work with students at different levels of statistical expertise and have shaped my approach to teaching. Moreover, I am engaged in freelance teaching, where I share my expertise in statistics with bachelors' and masters' students.

TEACHING AND TEACHING ASSISTANTSHIP FOR STATISTICS COURSES

Spring 2024	Fundamentals of Biostatistics using IBM SPSS (UPES Dehradun)
Spring 2024	Biostatistical Methods in Clinical Research using IBM SPSS (UPES Dehradun)
Spring 2021	SI 534 Nonparametric statistics (Teaching Assistant, IIT Bombay)

Autumn 2021	SI 505 Multivariate analysis (Teaching Assistant, IIT Bombay)
Spring 2020	SI 422 Regression Analysis (Teaching Assistant, IIT Bombay)
Autumn 2020	SI 417 Introduction to Probability Theory (Teaching Assistant, IIT Bombay)
Spring 2019	SI 404 Applied Stochastic Process (Teaching Assistant, IIT Bombay)

ACADEMIC ACTIVITIES

1. Participated in the Workshop on Quantile Regression organized by International Indian Statistical Association (IISA) at IIT Bombay, 2019.
2. Participated in the R-Conference organized by International Indian Statistical Association (IISA) at IIT Bombay, 2019.
3. Participated in the “IFCAM Summer School Workshop on Mathematics For Data Science” organized by IISc Bangalore, 2019.
4. Participated in the “Scientific writing using LATEX” organized by Dr. Shakuntala Misra National Rehabilitation University, Lucknow, 2017.
5. Completed certification course “Introduction to Data Analytics”, IIT Madras through NPTEL.
6. Completed certification course “Introduction to Software R”, IIT Kanpur through NPTEL.

DEPARTMENTAL RESPONSIBILITIES

2012 - 2013	Leaded the Registration team of 32th Mathematics Olympiad 2013, Department of Mathematics, Indian Institute of Technology, Bombay.
2012 - 2013	Member, Design and Logistic team of 32th Mathematics Olympiad 2013, Department of Mathematics, Indian Institute of Technology, Bombay.

TECHNICAL STRENGTHS

Computer Languages	R, IBM SPSS, MATLAB, Mathematica, Python (Basic Level)
Software & Tools	LaTeX, MS Office, Excel

ACADEMIC ACCOMPLISHMENTS

1. Got selected for National Board for Higher Mathematics (NBHM) Postdoctoral Fellowship, 2023(II).
2. Achieved All India Rank 72 in GATE (Statistics) 2020.
3. Achieved All India Rank 67 in GATE (Statistics) 2019.
4. Participated in the Workshop on Quantile Regression organized by IISA at IIT Bombay, 2019.
5. Participated in the R-Conference organized by IISA at IIT Bombay, 2019.
6. Achieved All India Rank 325 in GATE (Mathematics) 2017.
7. Qualified Indian Statistical Service (ISS) Examination - 2016, 2017 conducted by Union Public Service Commission (UPSC).
8. Qualified RBI Grade B officer Examination - 2018, 2019, 2021 conducted by Reserve Bank of India Selection Board (RBISB).
9. Got selected for summer intenship from Ministry of Statistics and Programme Implementation (MoS&PI) in 2012-13.

10. Achieved All India Rank (AIR) 85 in IIT JAM Statistics 2011.
11. Achieved All India Rank (AIR) 90 in IIT JAM Statistics 2009.

REFERENCES

Prof. P. Vellaisamy

Department of Mathematics
Indian Institute of Technology, Bombay
Mumbai, Maharashtra - 400076
Email: pv@math.iitb.ac.in

Prof. Siuli Mukhopadhyay

Department of Mathematics
Indian Institute of Technology, Bombay
Mumbai, Maharashtra - 400076
Email: siuli@math.iitb.ac.in

Prof. Mohd. Arshad

Department of Mathematics
Indian Institute of Technology, Indore
Simrol, Khandwa Road, Indore 453552
Email: arshad@iiti.ac.in